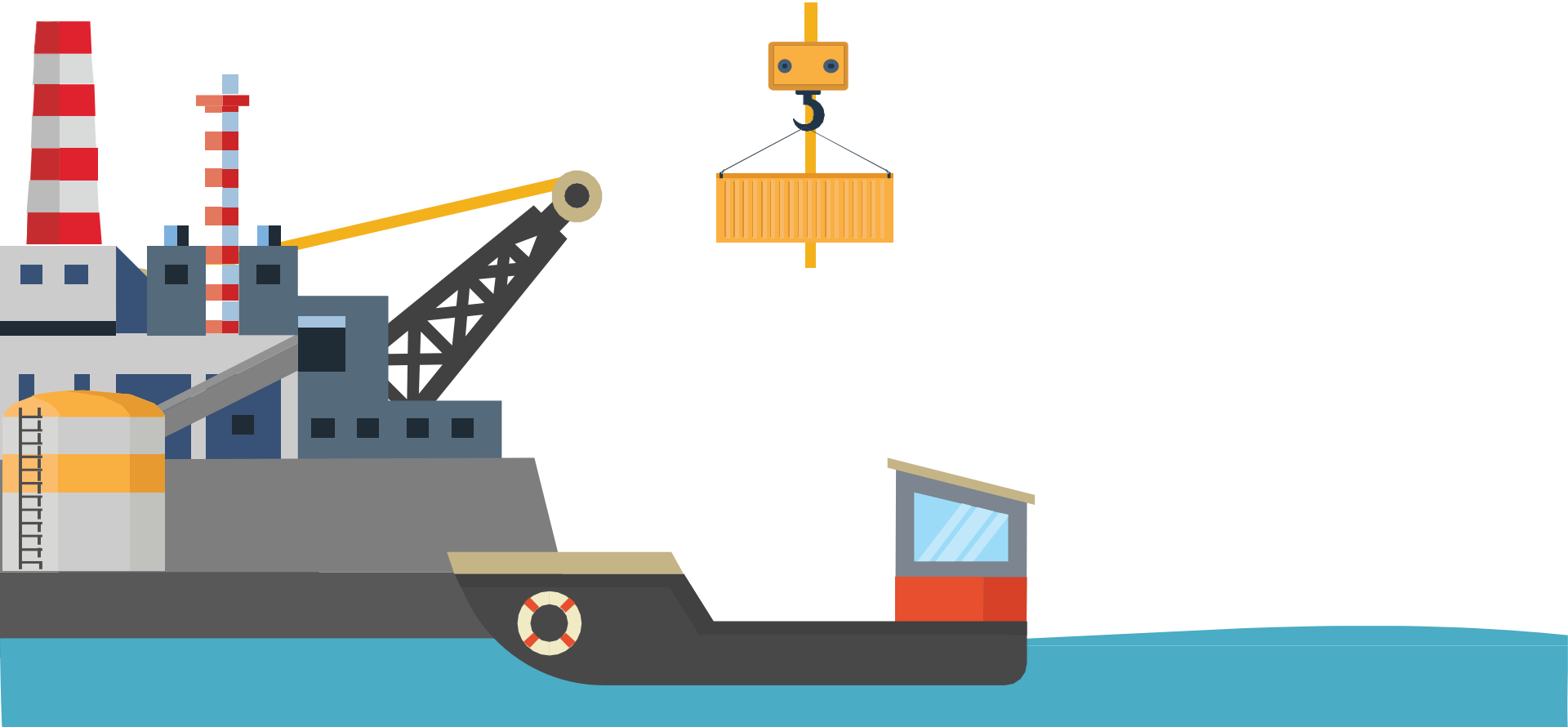
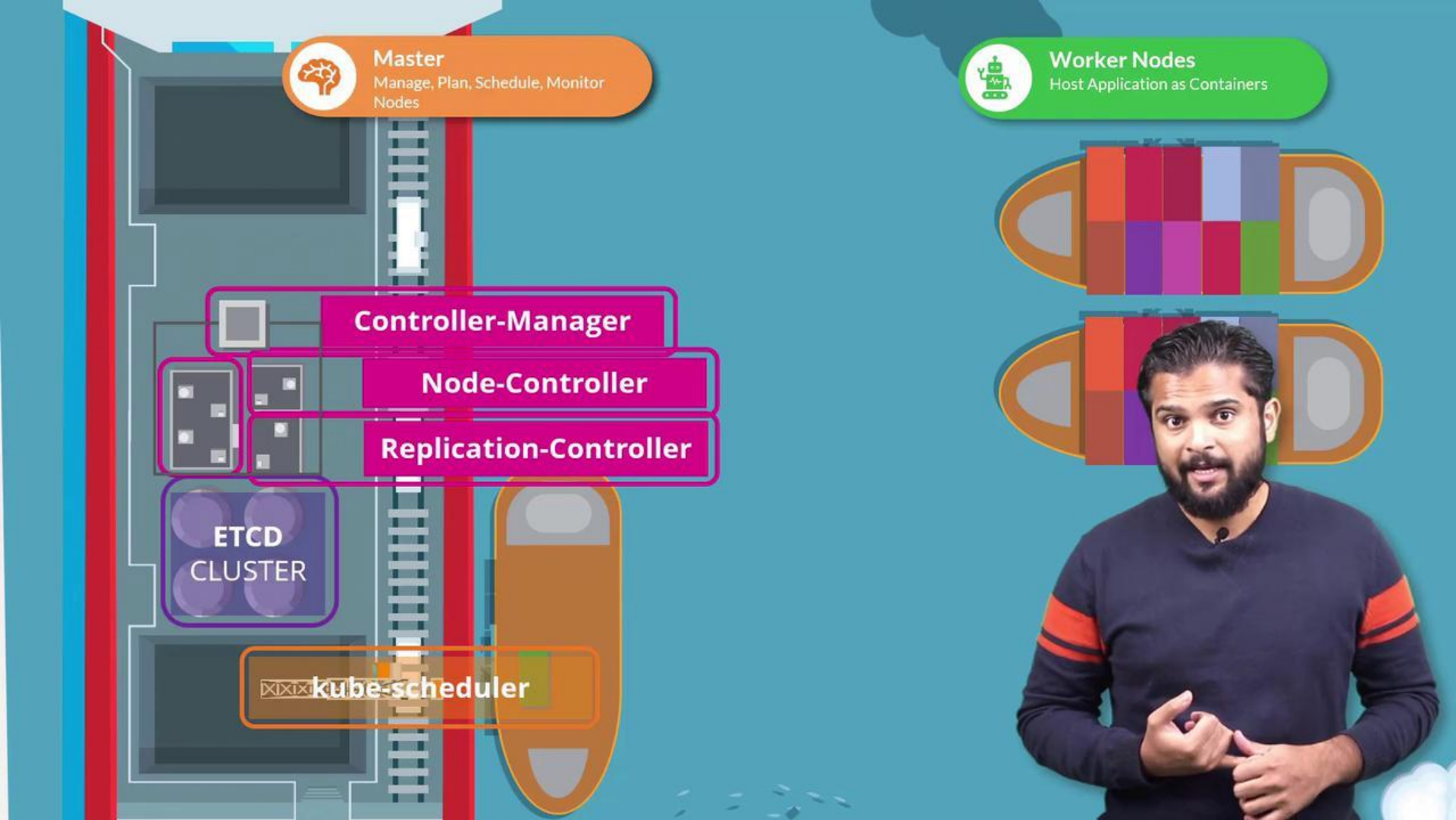
# Cluster Architecture

- ❑ Kubernetes Architecture
- ❑ ETCD For Beginners
- ❑ ETCD in Kubernetes
- ❑ Kube-API Server
- ❑ Controller Managers
- ❑ Kube Scheduler
- ❑ Kubelet
- ❑ Kube Proxy

# KUBERNETES ARCHITECTURE



Master
Manage, Plan, Schedule, Monitor Nodes
Worker Nodes
Host Application as Containers


Master
Manage, Plan, Schedule, Monitor Nodes
Worker Nodes
Host Application as Containers
Controller-Manager
Node-Controller
Replication-Controller
ETCD
CLUSTER
kube-scheduler

# Kubernetes Architecture

**Master**
Manage, Plan, Schedule, Monitor Nodes

ETCD CLUSTER

kube-apiserver

Kube Controller Manager

kube-scheduler

**Worker Nodes**
Host Application as Containers

kubelet

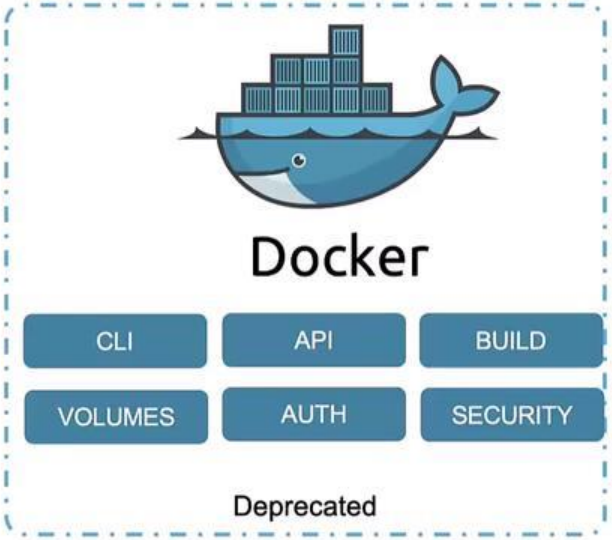
Kube-proxy

Container Runtime Engine
Run containers
docker rkt

kubelet

Kube-proxy

Container Runtime Engine
Run containers
docker rkt




Container Runtime Interface (CRI)

# ETCD

## Para principiantes

# Objectives

- What is ETCD?
- What is a Key-Value Store?
- How to get started quickly?
- How to operate ETCD?
- What is a distributed system?
- How ETCD Operates
- RAFT Protocol
- Best practices on number of nodes

ETCD is a distributed reliable key-value store that is Simple, Secure & Fast

# key-value store

Tabular/Relational Databases

| Name | Age | Location | Salary | Grade |
|---|---|---|---|---|
| John Doe | 45 | New York | 5000 | |
| Dave Smith | 34 | New York | 4000 | |
| Aryan Kumar | 10 | New York | | A |
| Lauren Rob | 13 | Bangalore | | C |
| Lily Oliver | 15 | Bangalore | | B |

# key-value store

| Key | Value |
|---|---|
| Name | John Doe |
| Age | 45 |
| Location | New York |
| Salary | 5000 |

| Key | Value |
|---|---|
| Name | Dave Smith |
| Age | 34 |
| Location | New York |
| Salary | 4000 |
| Organization | ACME |

| Key | Value |
|---|---|
| Name | Aryan Kumar |
| Age | 10 |
| Location | New York |
| Grade | A |

| Key | Value |
|---|---|
| Name | Lauren Rob |
| Age | 13 |
| Location | Bangalore |
| Grade | C |

| Key | Value |
|---|---|
| Name | Lily Oliver |
| Age | 15 |
| Location | Bangalore |
| Grade | B |

# key-value store

```
{
  "name": "John Doe",
  "age": 45,
  "location": "New York",
  "salary": 5000
}
```

```
{
  "name": "Dave Smith",
  "age": 34,
  "location": "New York",
  "salary": 4000,
  "organization": "ACME"
}
```

```
{
  "name": "Aryan Kumar",
  "age": 10,
  "location": "New York",
  "Grade": "A"
}
```

```
{
  "name": "Lily Oliver",
  "age": 15,
  "location": "Bangalore",
  "Grade": "B"
}
```

```
{
  "name": "Lauren Rob",
  "age": 13,
  "location": "Bangalore",
  "Grade": "C"
}
```

# Operate ETCD

## 3. Run ETCD Service

```
./etcd
```

```
./etcdctl set key1 value1
```

```
./etcdctl get key1
value1
```

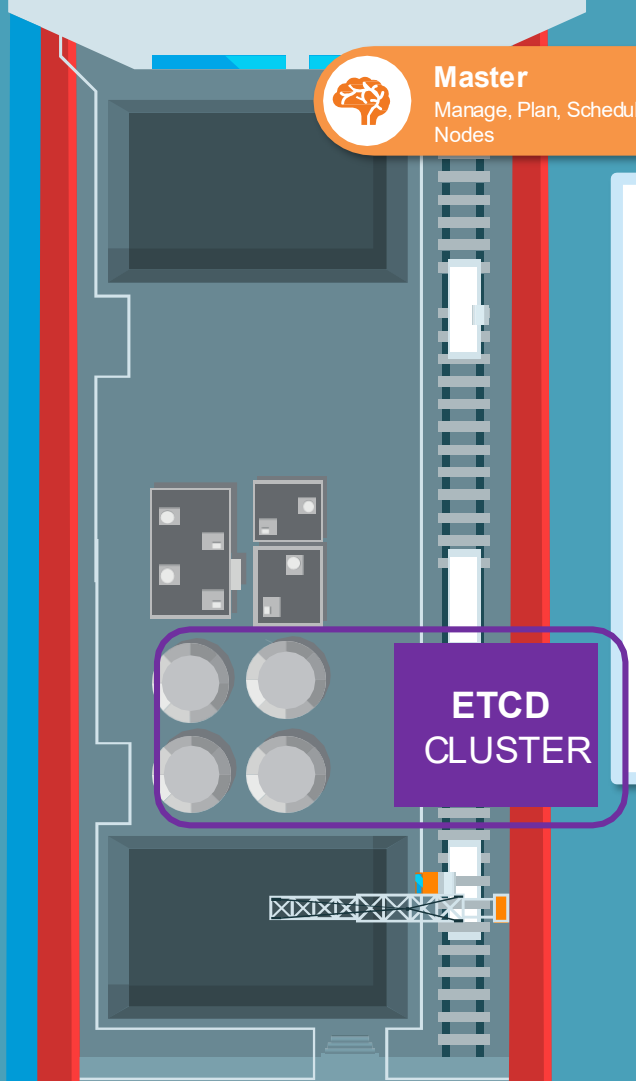
```
./etcdctl
NAME:
    etcdctl - A simple command line client for etcd.

COMMANDS:
    backup          backup an etcd directory
    cluster-health  check the health of the etcd cluster mk
                    make a new key with a given value
    mkdir           make a new directory
    rm              remove a key or a directory
    rmdir           removes the key if it is an empty directory or a key-value pair get        retrieve the value
    of a key
```

# ETCD Versions

| | | |
|---|---|---|
| v0.1 | Aug 2013 | |
| v0.5 | Dec 2014 | |
| v2.0 | Feb 2015 | |
| v3.1 | Jan 2017 | |
| | Nov 2018 | CNCF Incubation |

# ETCD

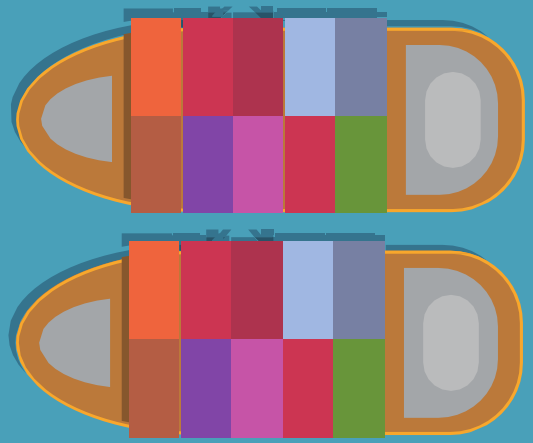
## En Kubernetes



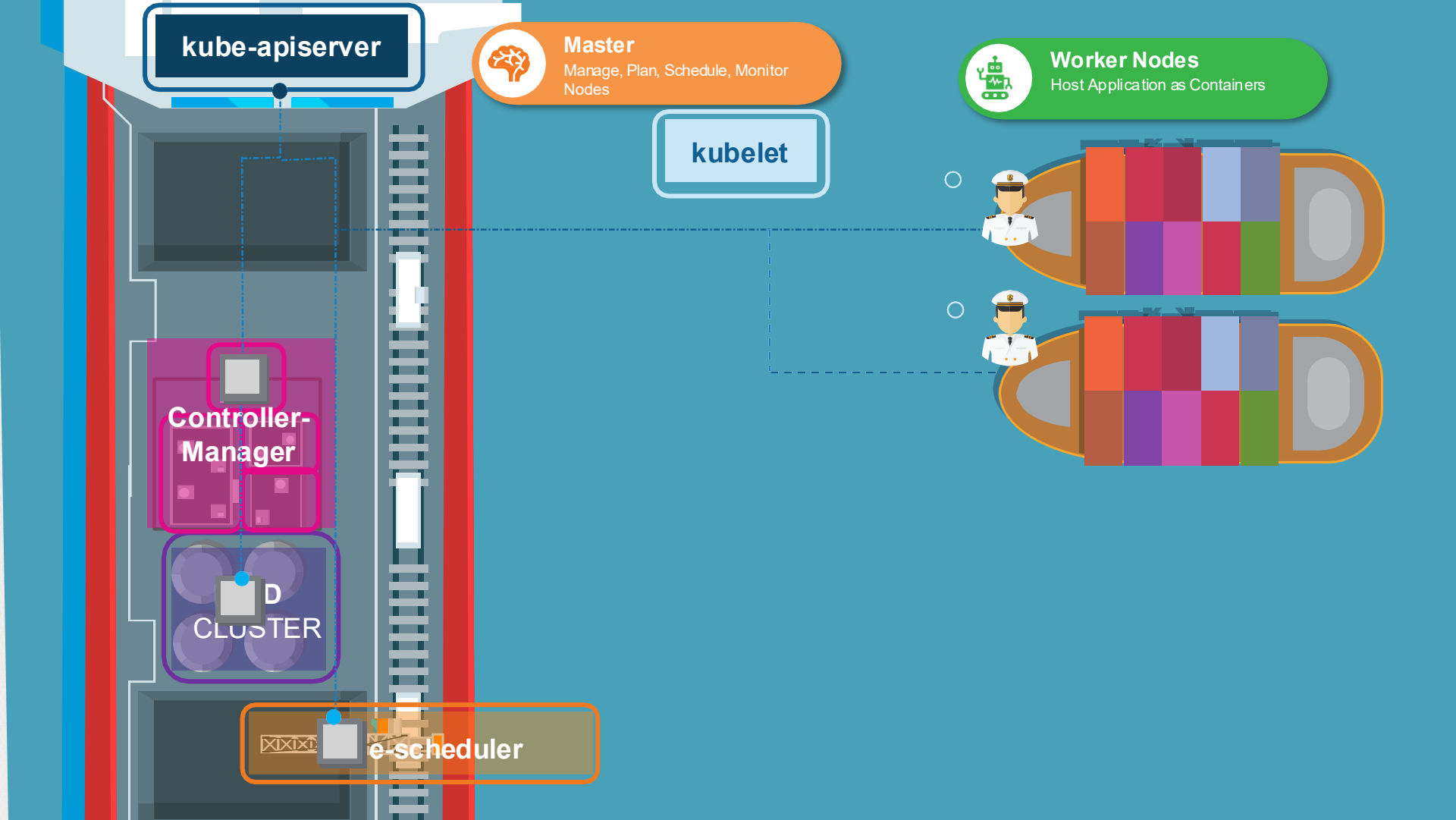

**Master**

Manage, Plan, Schedule, Monitor Nodes

- Nodes
- PODs
- Configs
- Secrets
- Accounts
- Roles
- Bindings
- Others



# kube-api server


kube-apiserver
Master
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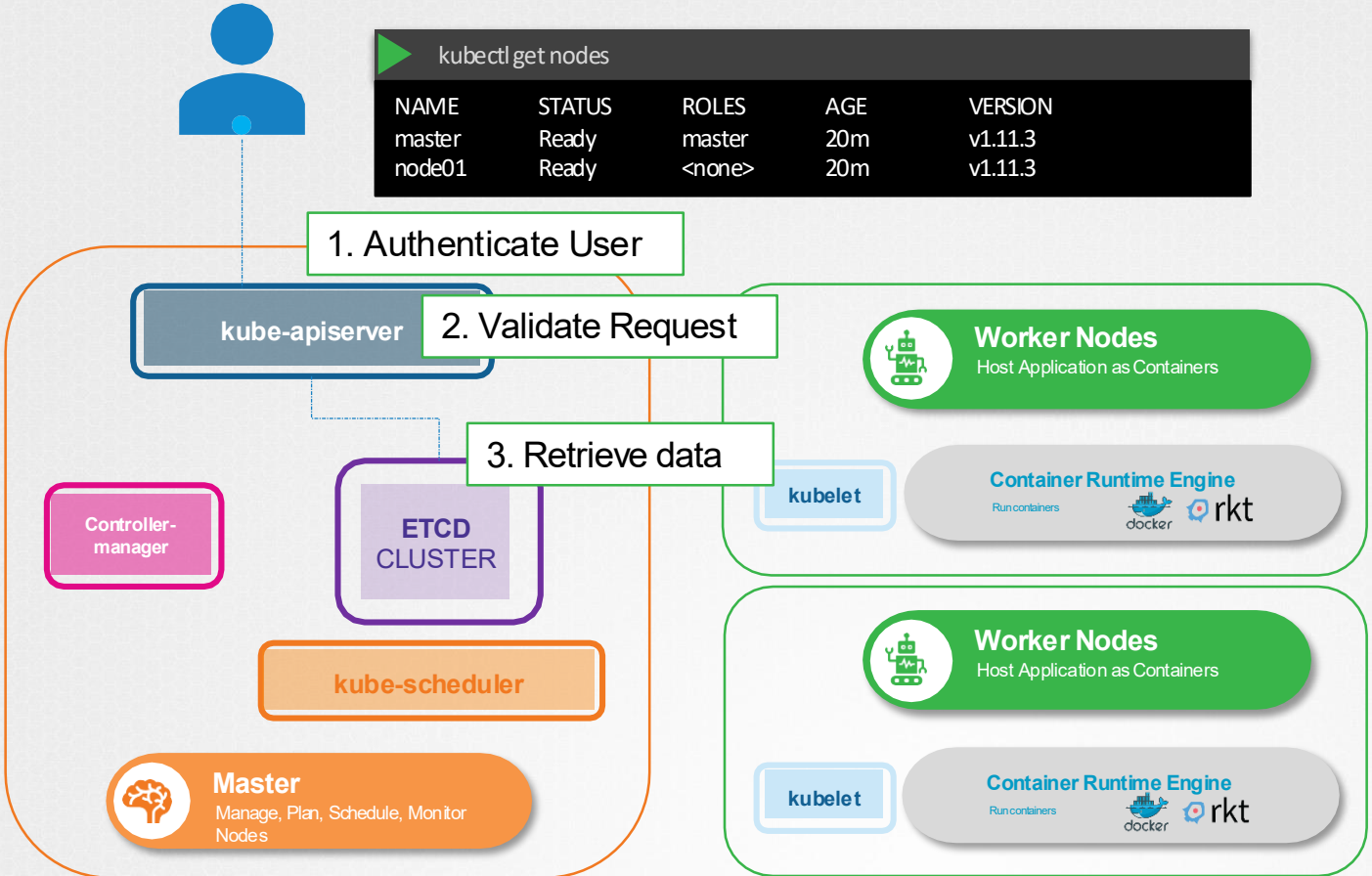
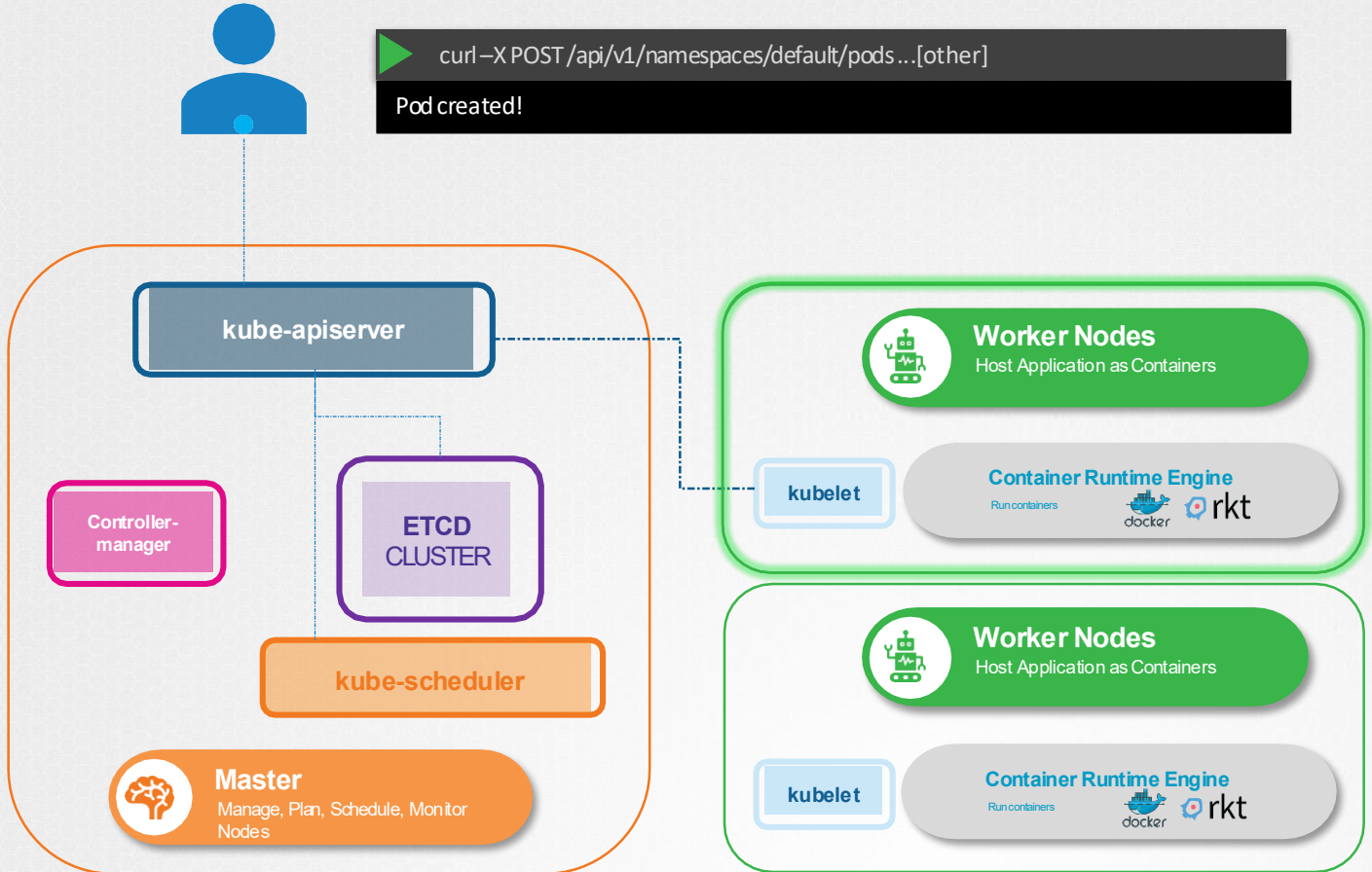
Worker Nodes
Host Application as Containers
kubelet
Controller-Manager
D
CLUSTER
e-scheduler

# Kubernetes Architecture



# Kubernetes Architecture




1. Authenticate User
2. Validate Request
3. Retrieve data
4. Update ETCD
5. Scheduler
6. Kubelet

# Kube-api Server

1. Authenticate User
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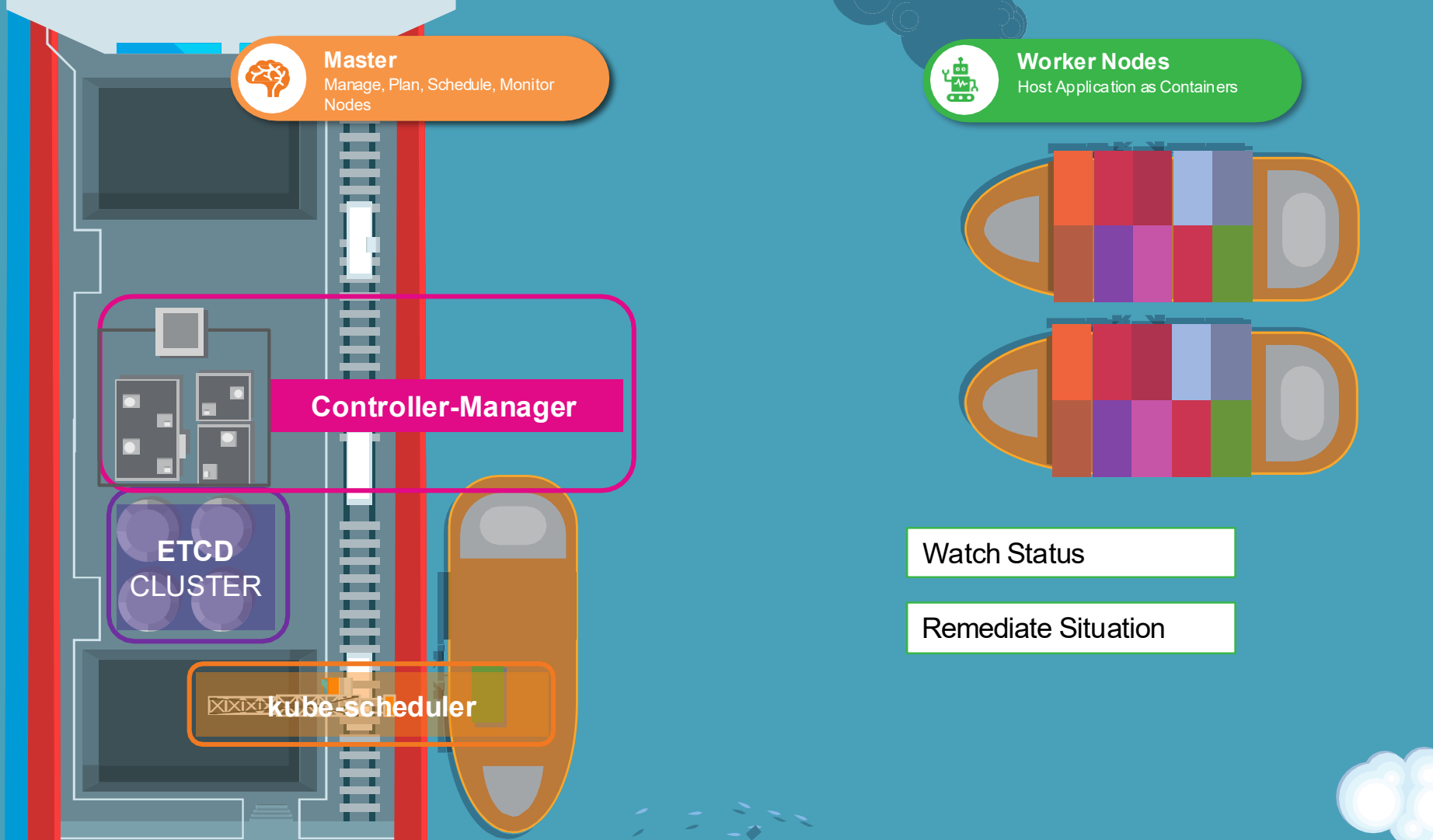
6. Kubelet

# Kube Controller Manager


Master
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ETCD
CLUSTER
kube-scheduler
Watch Status
Remediate Situation

# Controller

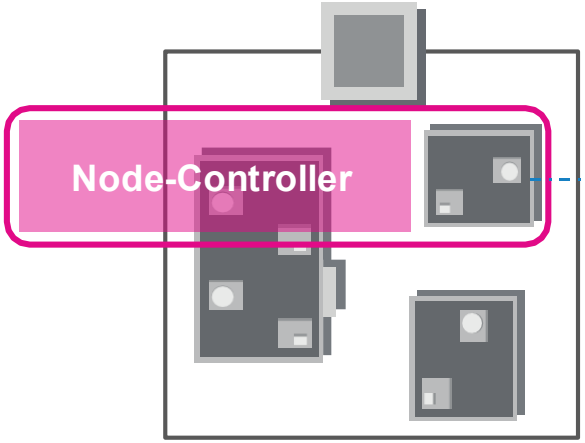
Watch Status

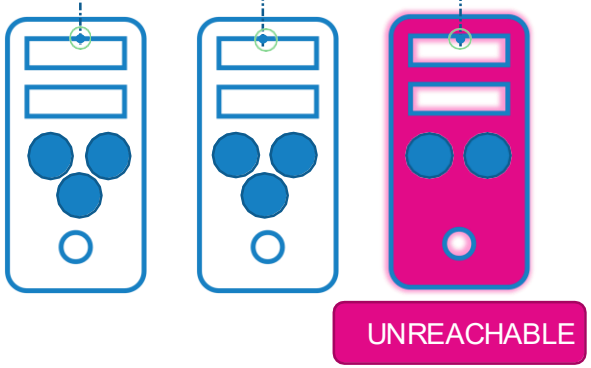
Remediate Situation

Node Monitor Period = 5s

Node Monitor Grace Period = 40s

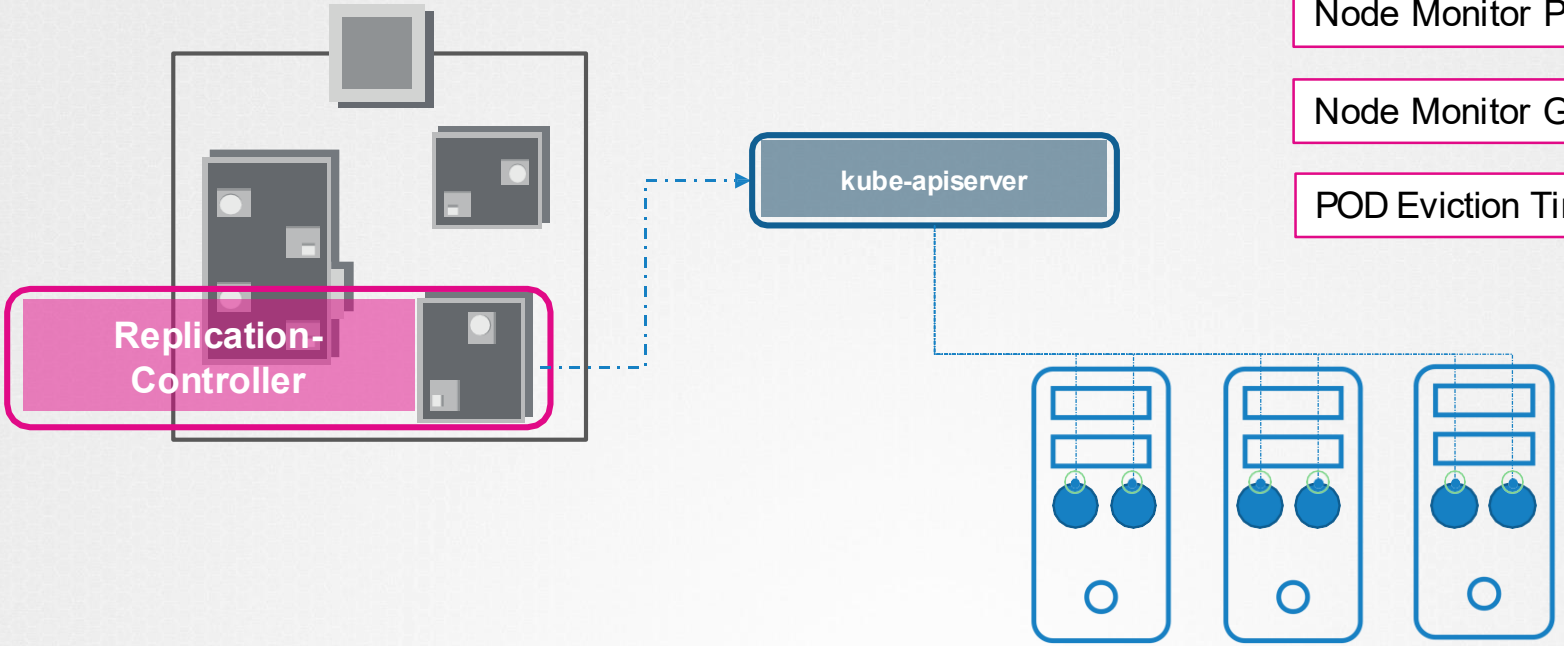
POD Eviction Timeout = 5m






```
kubectl get nodes
NAME      STATUS     ROLES    AGE   VERSION
worker-1  Ready      <none>   8d    v1.13.0
worker-2  NotReady   <none>   8d    v1.13.0
```

# Controller

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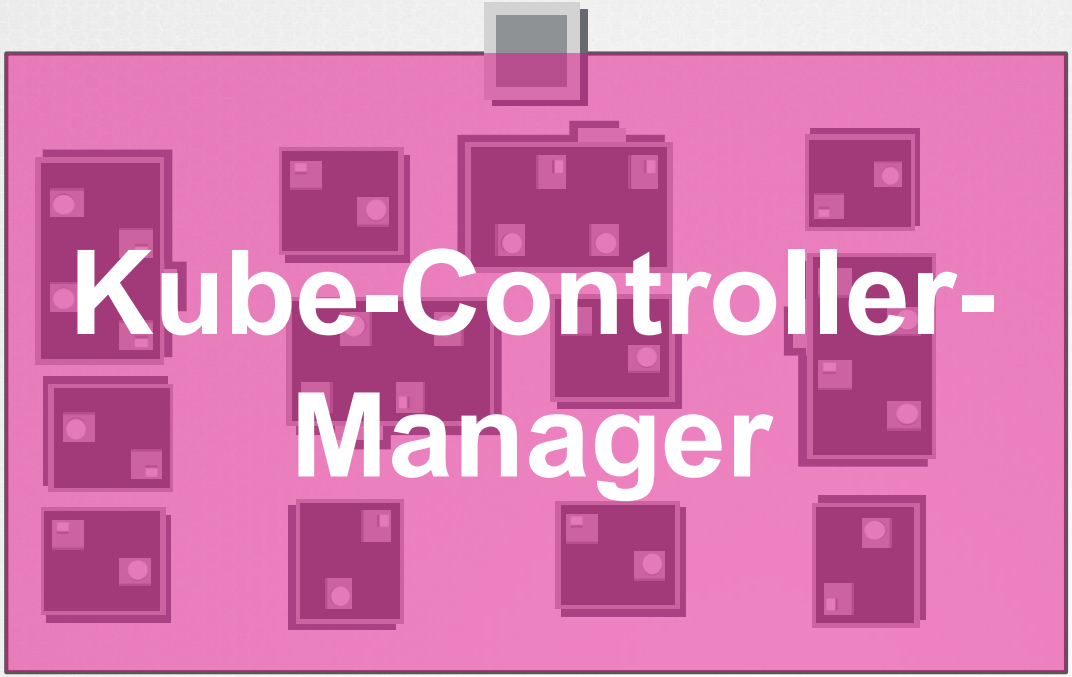


Watch Status

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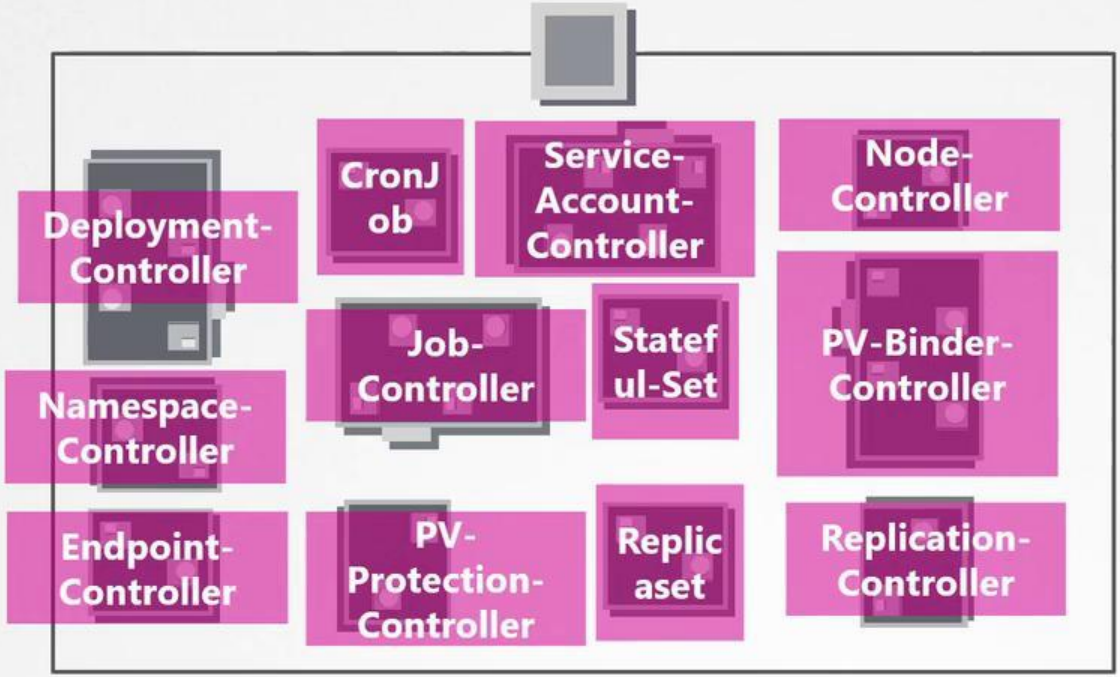


Watch Status

Remediate Situation

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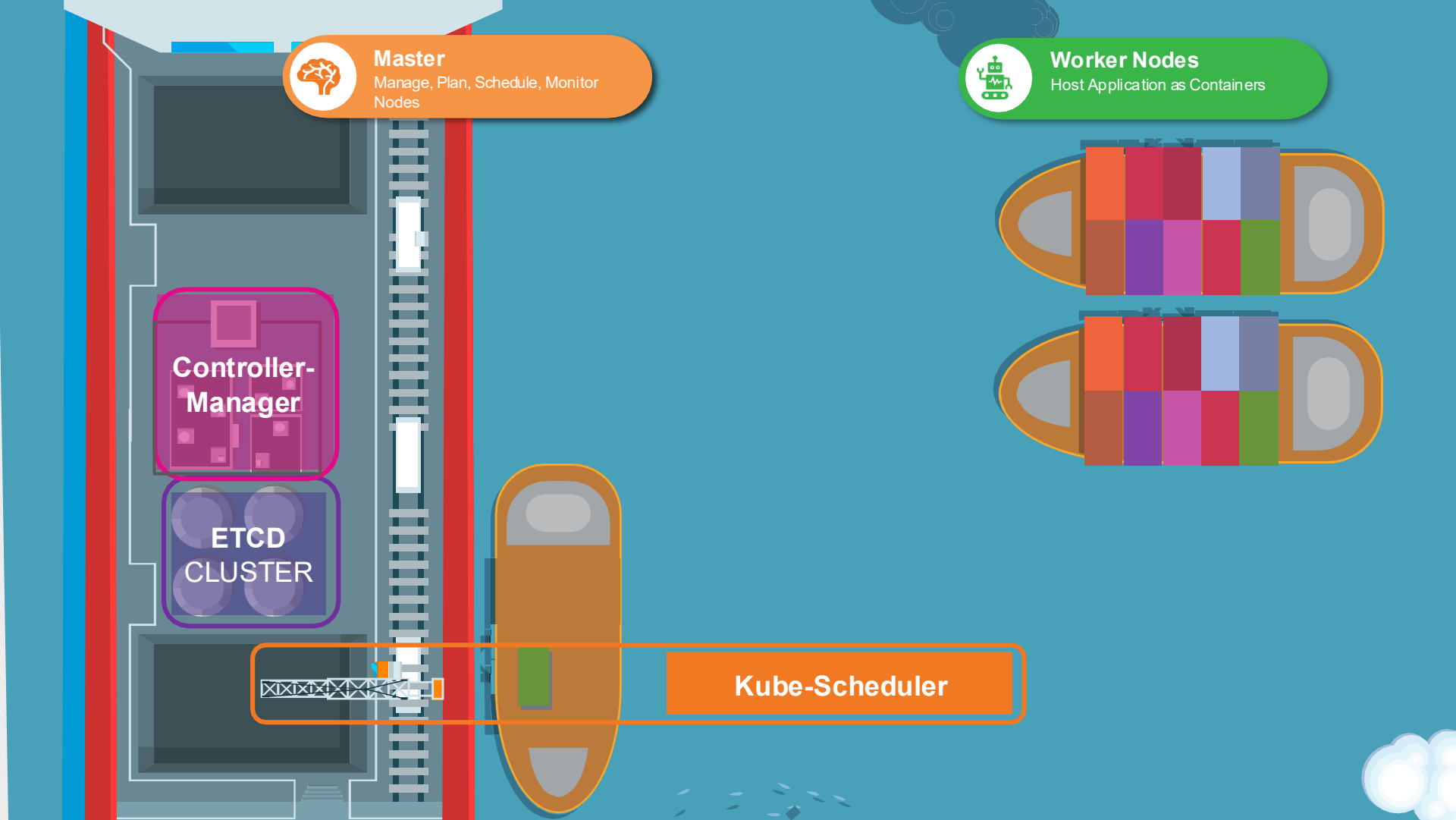
# Kube Scheduler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ETCD CLUSTER
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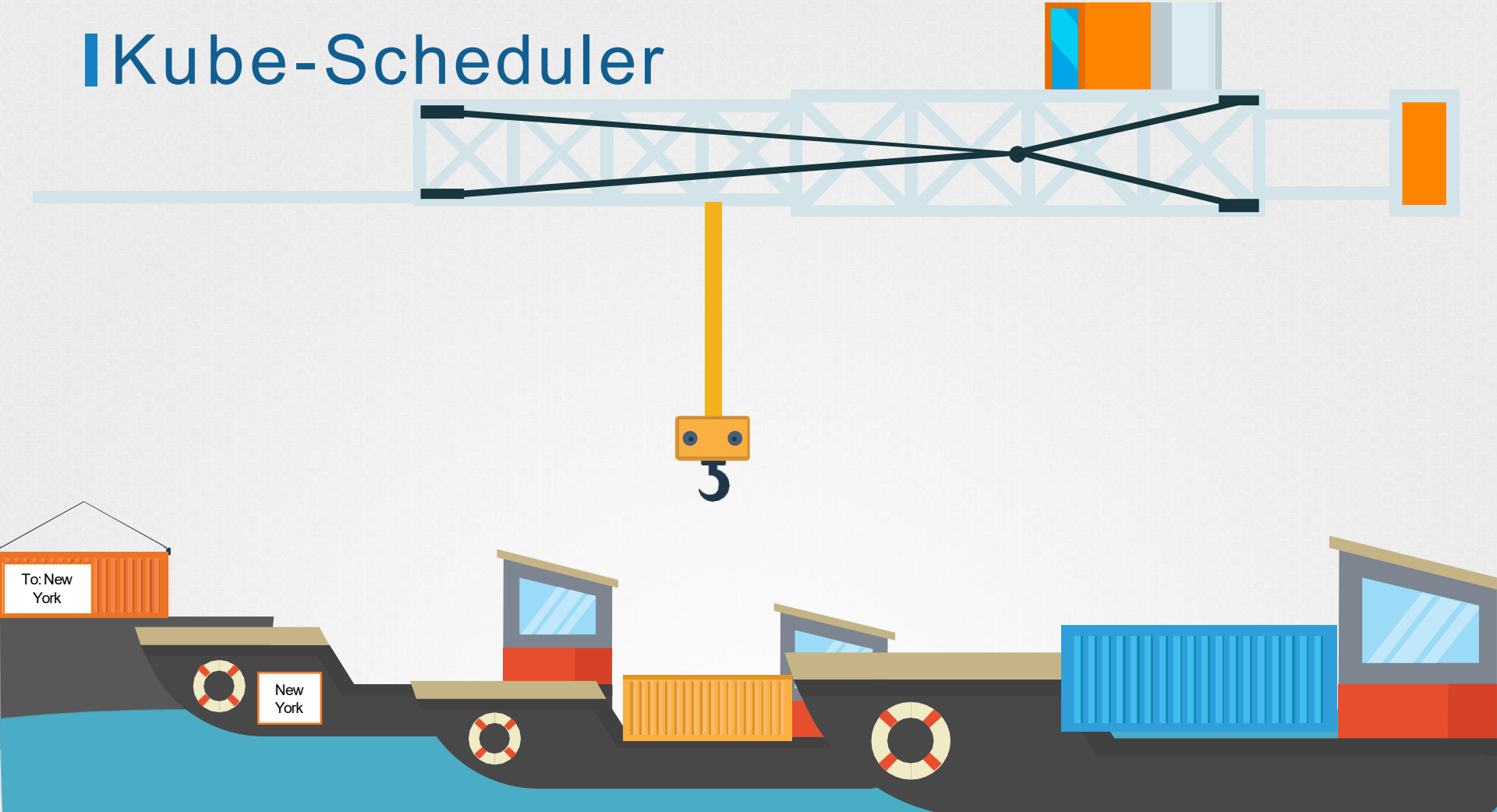
# Kube-Scheduler



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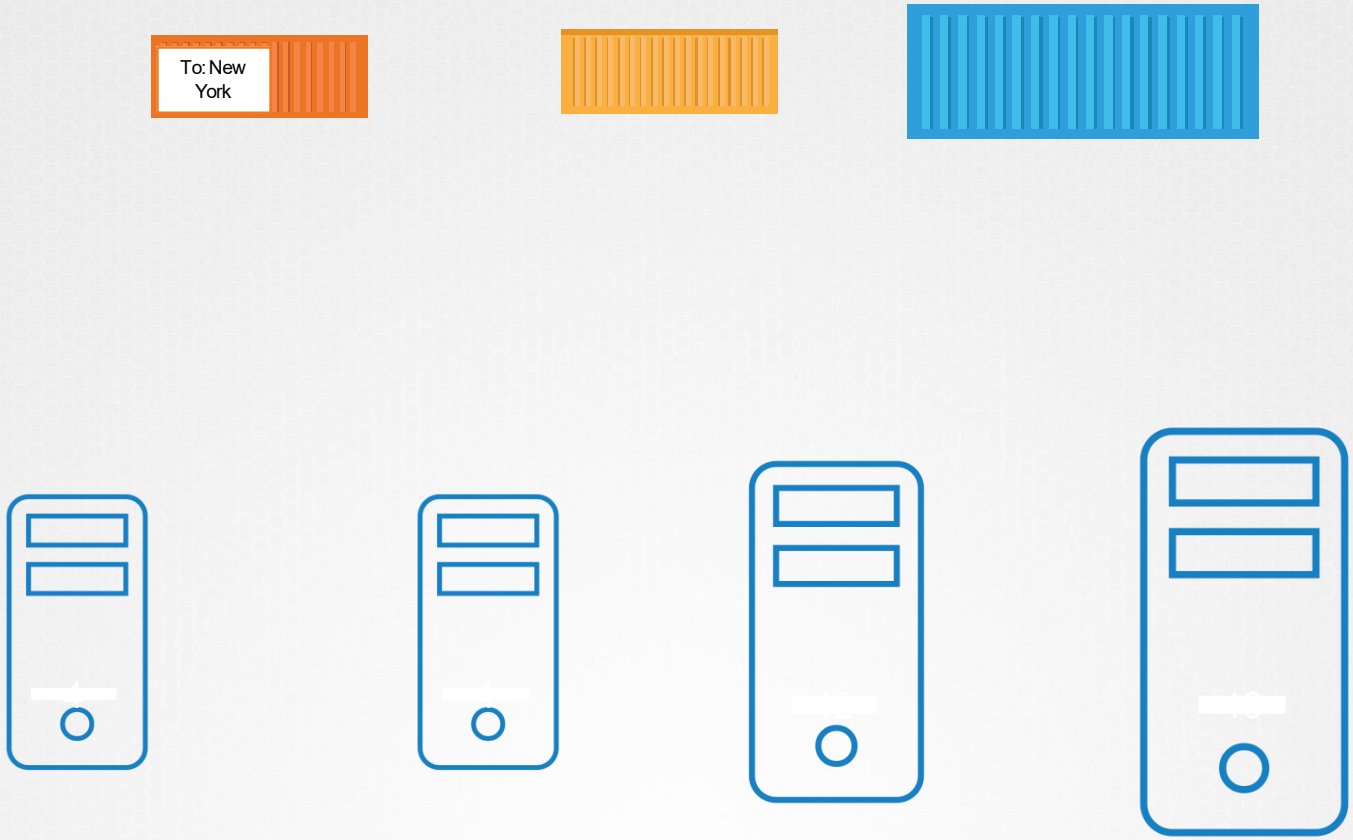
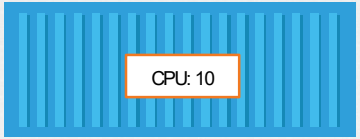
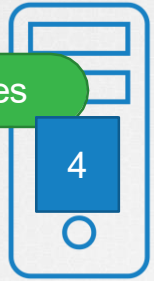


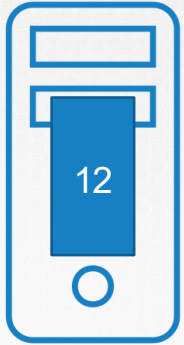
# Kube-Scheduler

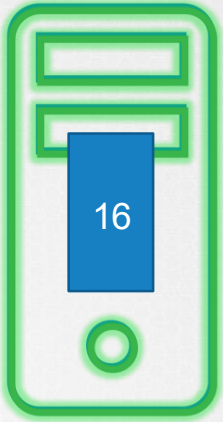



1. Filter Nodes

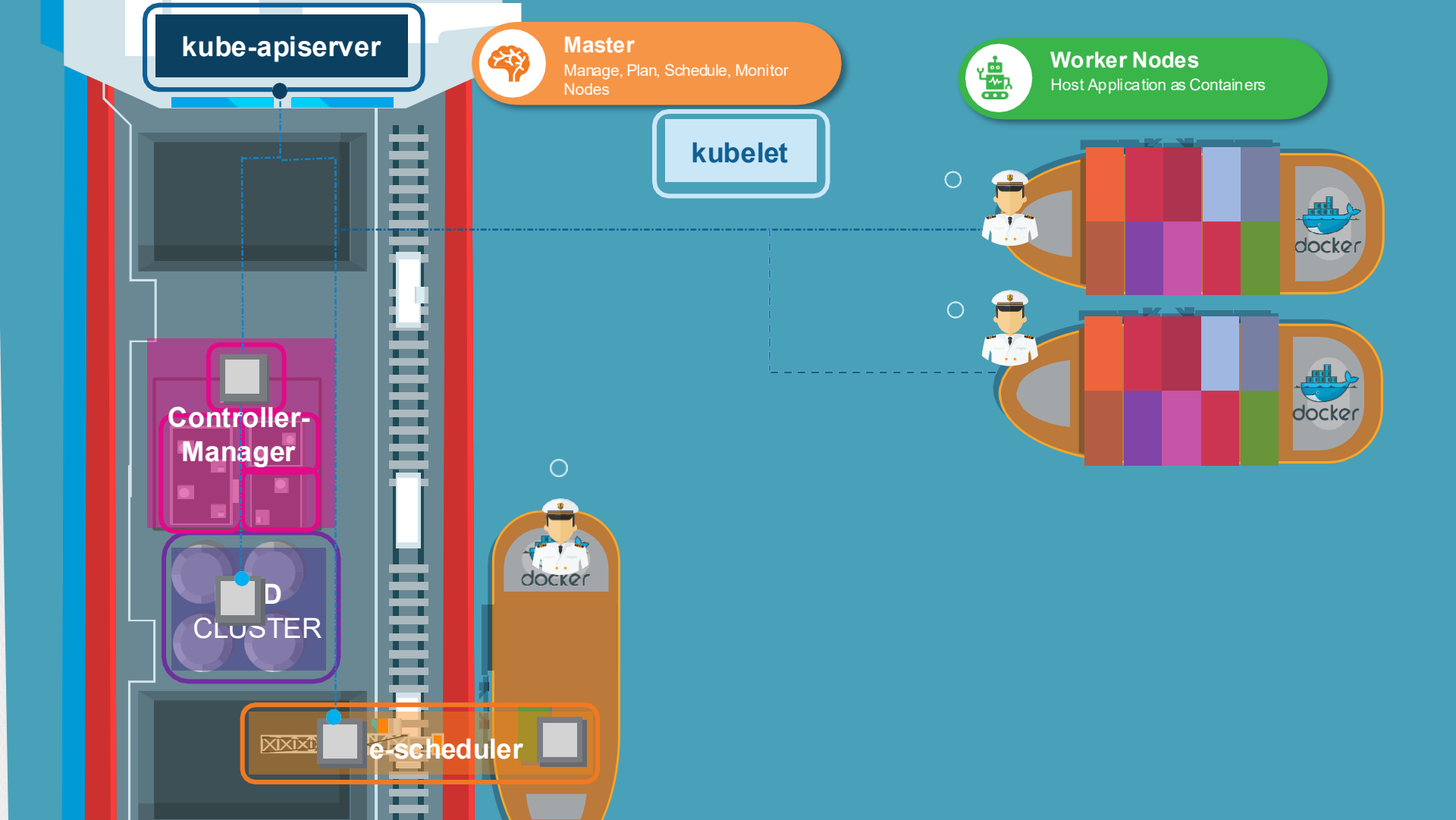
2. Rank Nodes





# Kubelet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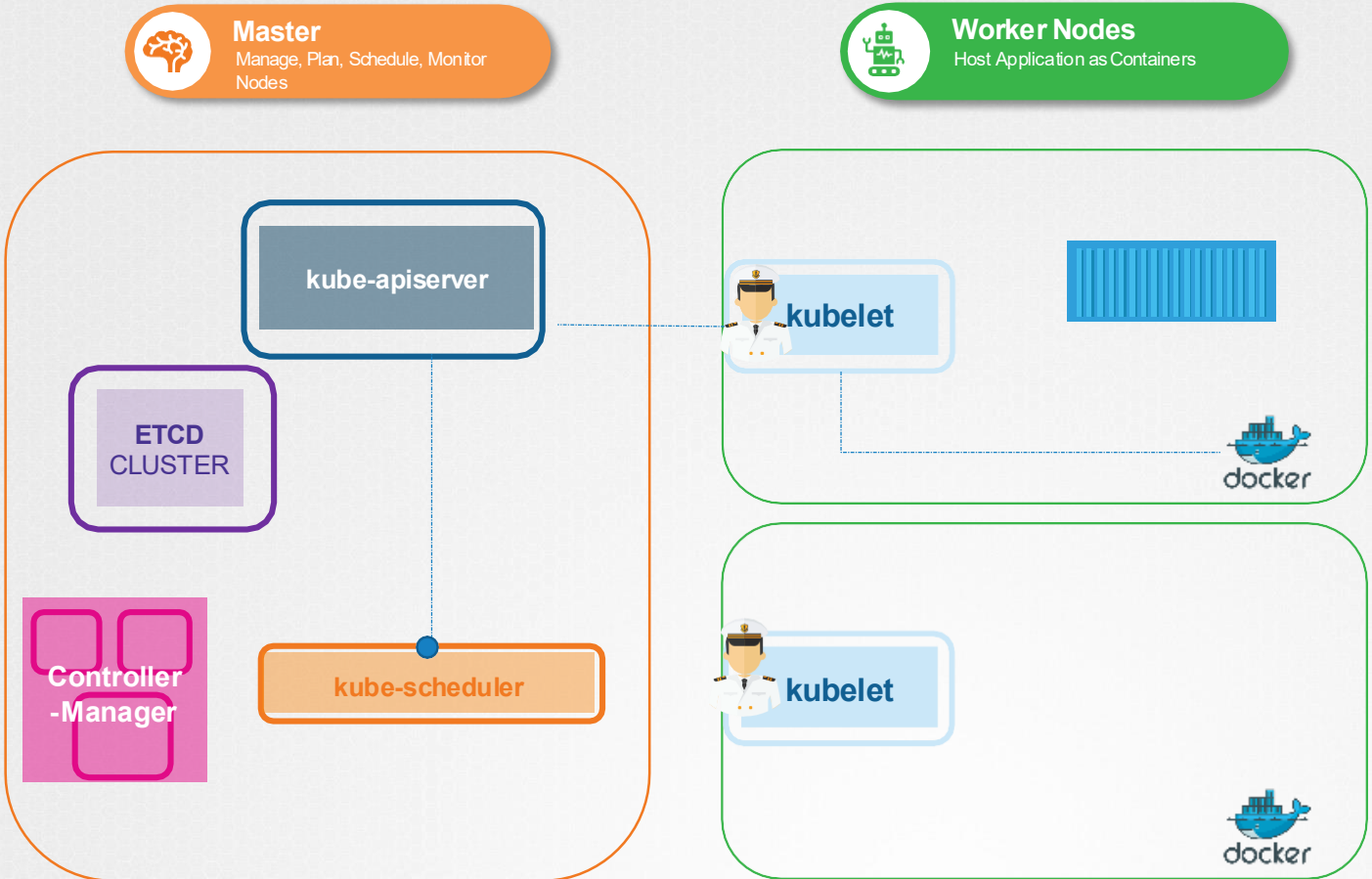


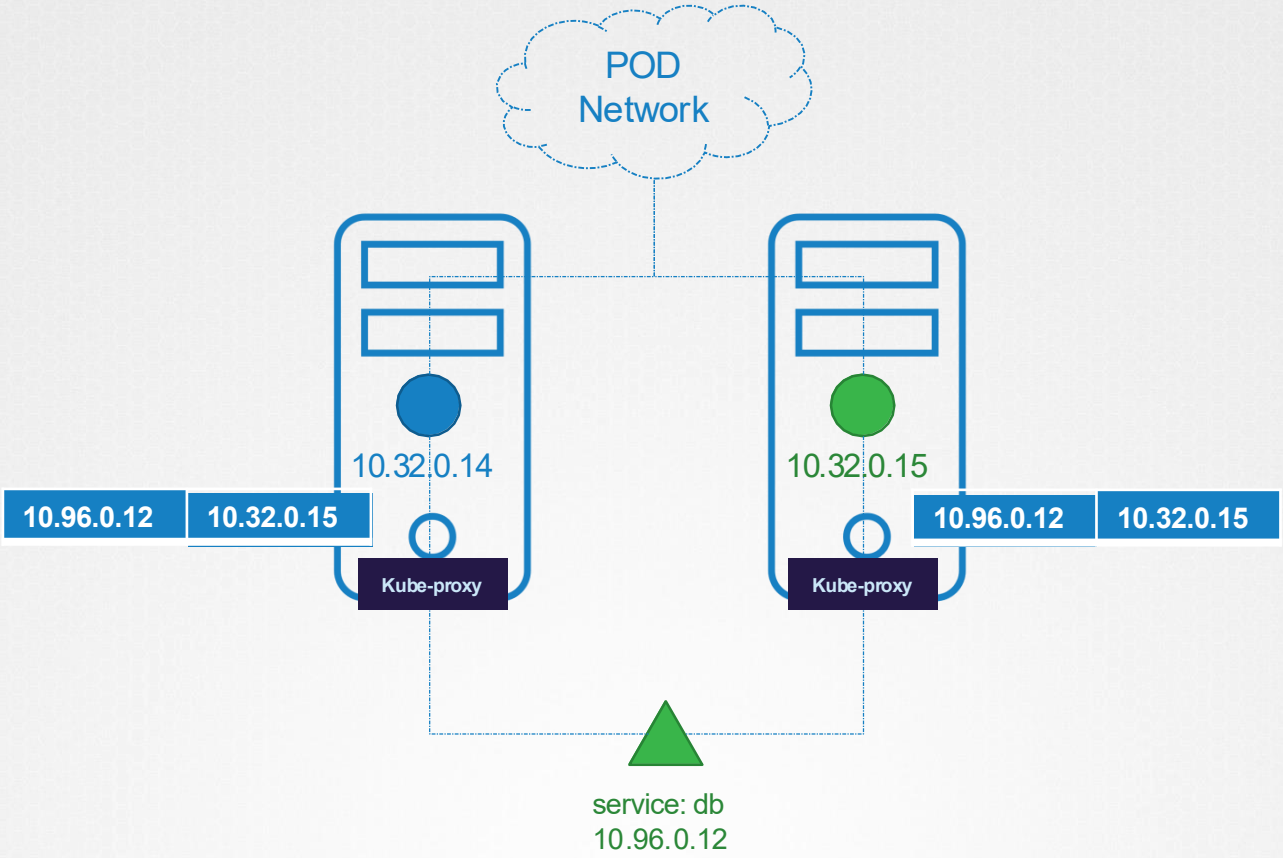

Register Node

Create PODs

Monitor Node & PODs

# Kube-proxy

# Kube-proxy


POD
Network
10.32.0.14
10.96.0.12
10.32.0.15
Kube-proxy
10.32.0.15
10.96.0.12
10.32.0.15
Kube-proxy
service: db
10.96.0.12